msvtx_chiSquared

overview · all regions and selections · rank 16 / 169

rank 16 / 169 by endcap / ht40 / D_equal

region	selection	D_equal	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg
endcap	all	0.486	0.576	1.000	0.643	0.620	0.635	0.603	0.582	34,970
endcap	ht40	0.485	0.565	1.000	0.631	0.609	0.629	0.592	0.571	7,390
endcap	bkg split	0.015	below 27,580 / above 7,390							
barrel	all	0.167	0.220	1.000	0.278	0.236	0.234	0.242	0.253	1,089
barrel	ht40	0.167	0.221	1.000	0.271	0.233	0.238	0.247	0.252	220
barrel	bkg split	0.035	below 869 / above 220							
all	all	0.405	0.490	1.000	0.550	0.525	0.544	0.509	0.494	36,059
all	ht40	0.399	0.479	1.000	0.539	0.514	0.533	0.498	0.484	7,610
all	bkg split	0.015	below 28,449 / above 7,610							

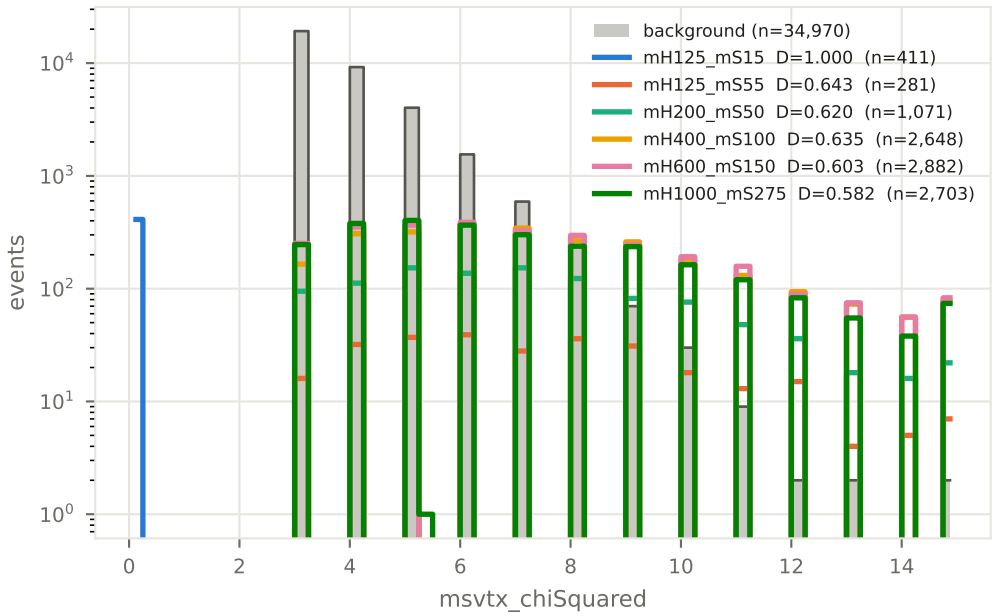
selections: all = every event; ht40 = htmissScalar_NOSYS >= 40 GeV, signal and background
pages: per region -> [all] 2x2, [ht40] 2x2, [background below vs above] 1x2

msvtx_chiSquared

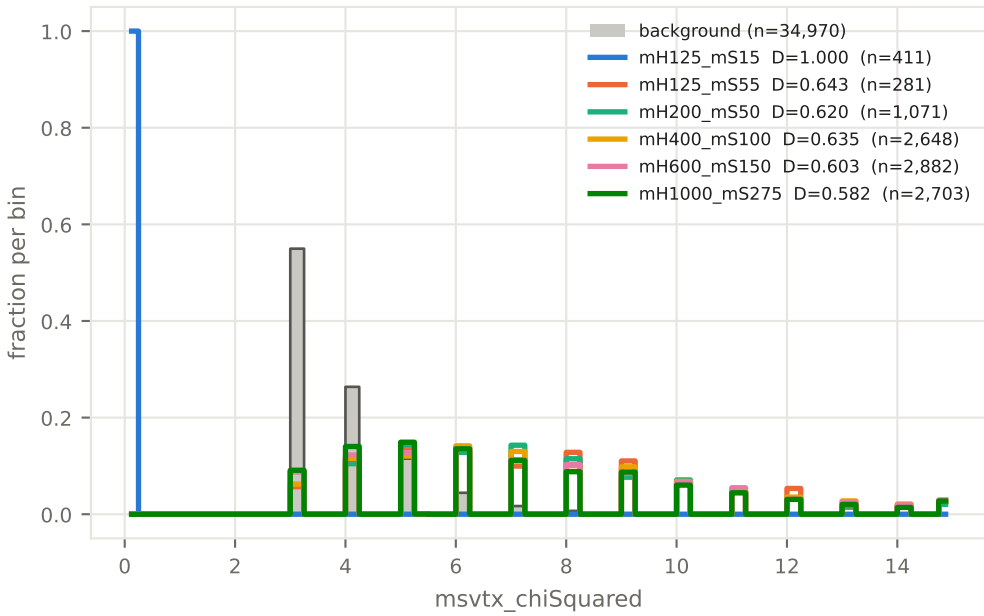
region: ENDCAP · selection: all (every event) · rank 16 / 169

D_equal 0.486 D_pooled 0.576 min 0.582 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 34,970 sig n = 9,996

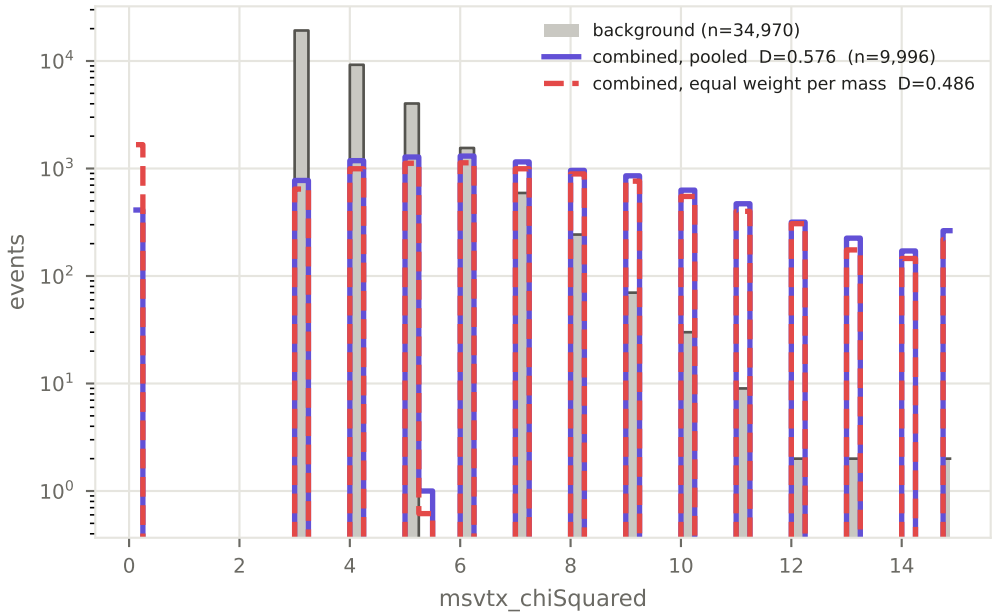
each mass point vs background · raw



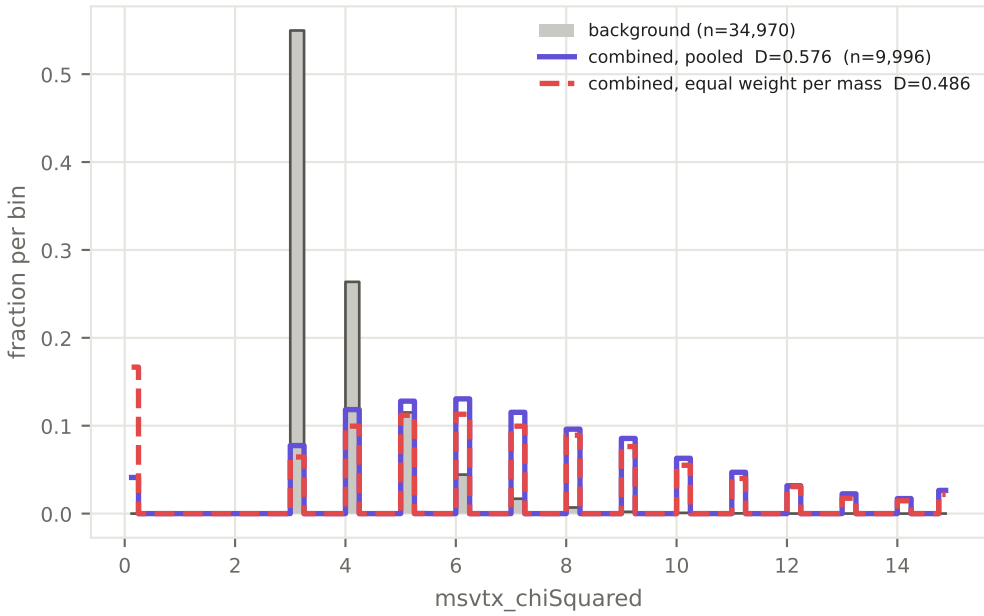
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



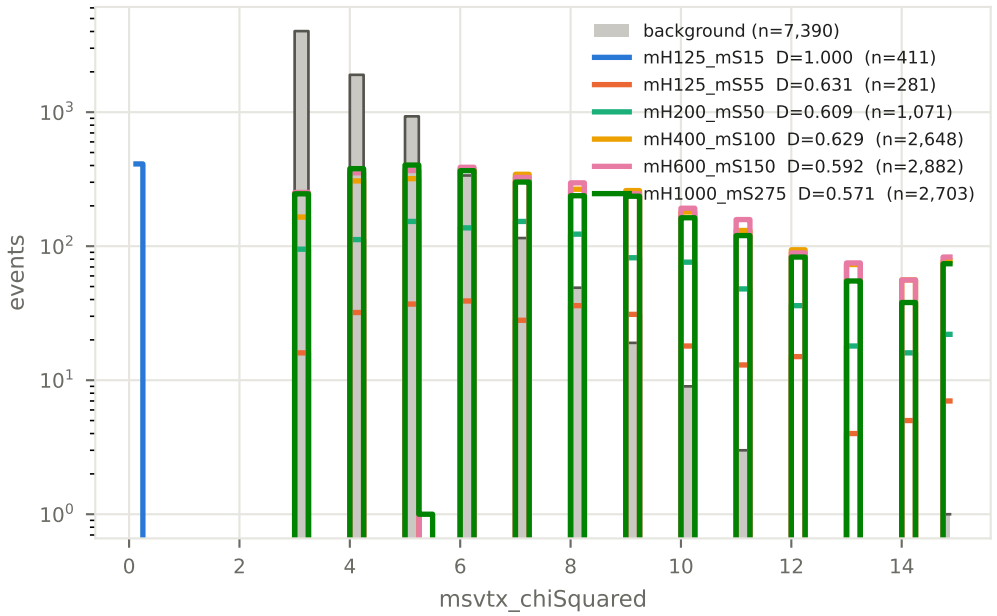
edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

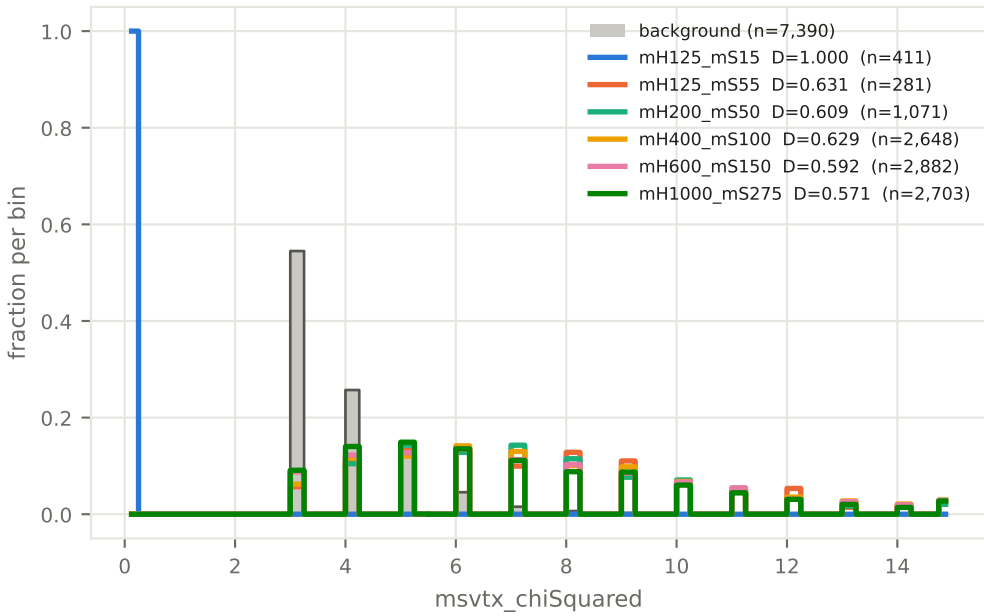
region: ENDCAP · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 16 / 169

D_equal 0.485 D_pooled 0.565 min 0.571 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 7,390 sig n = 9,996

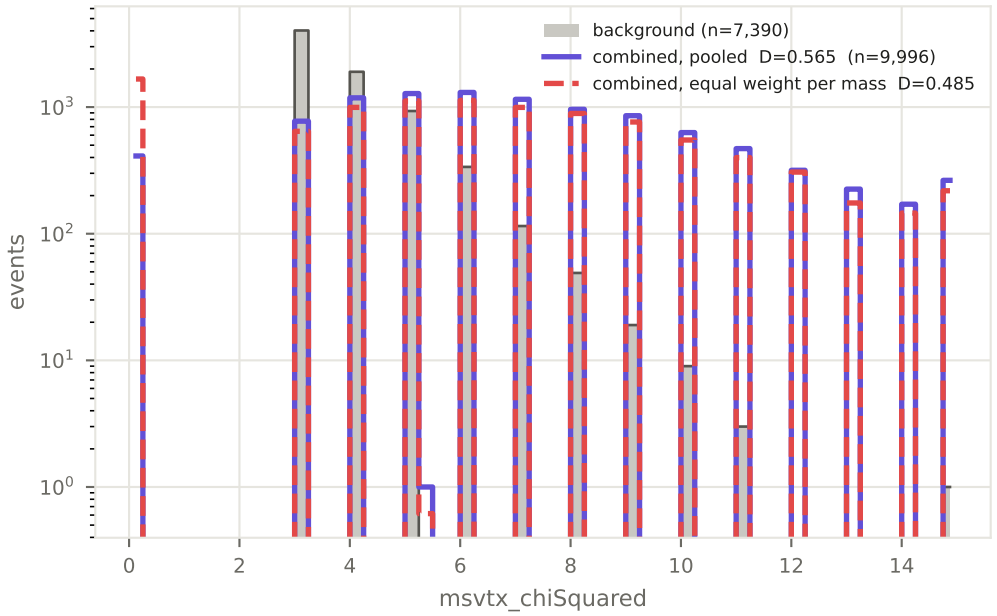
each mass point vs background · raw



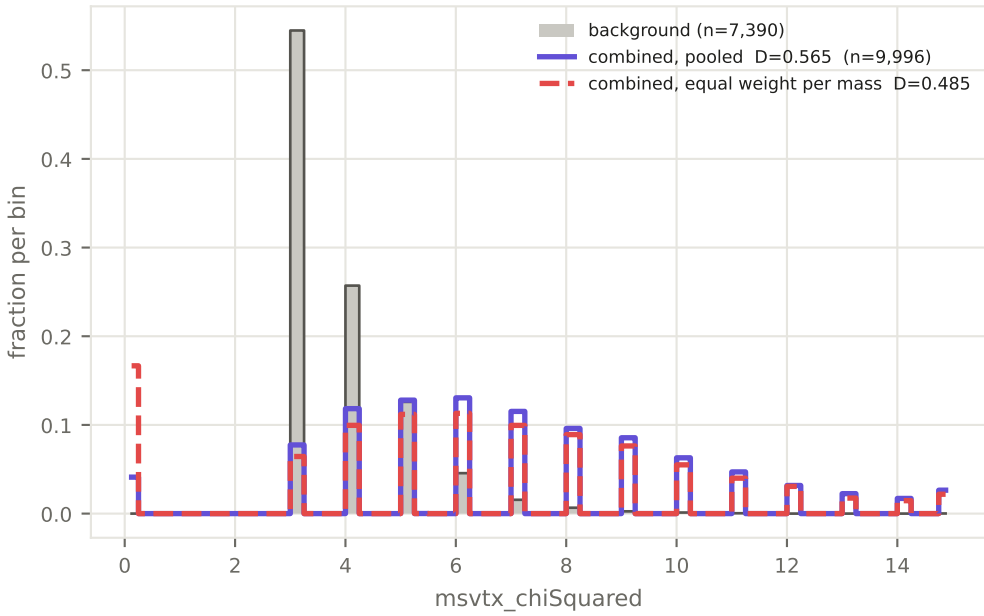
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

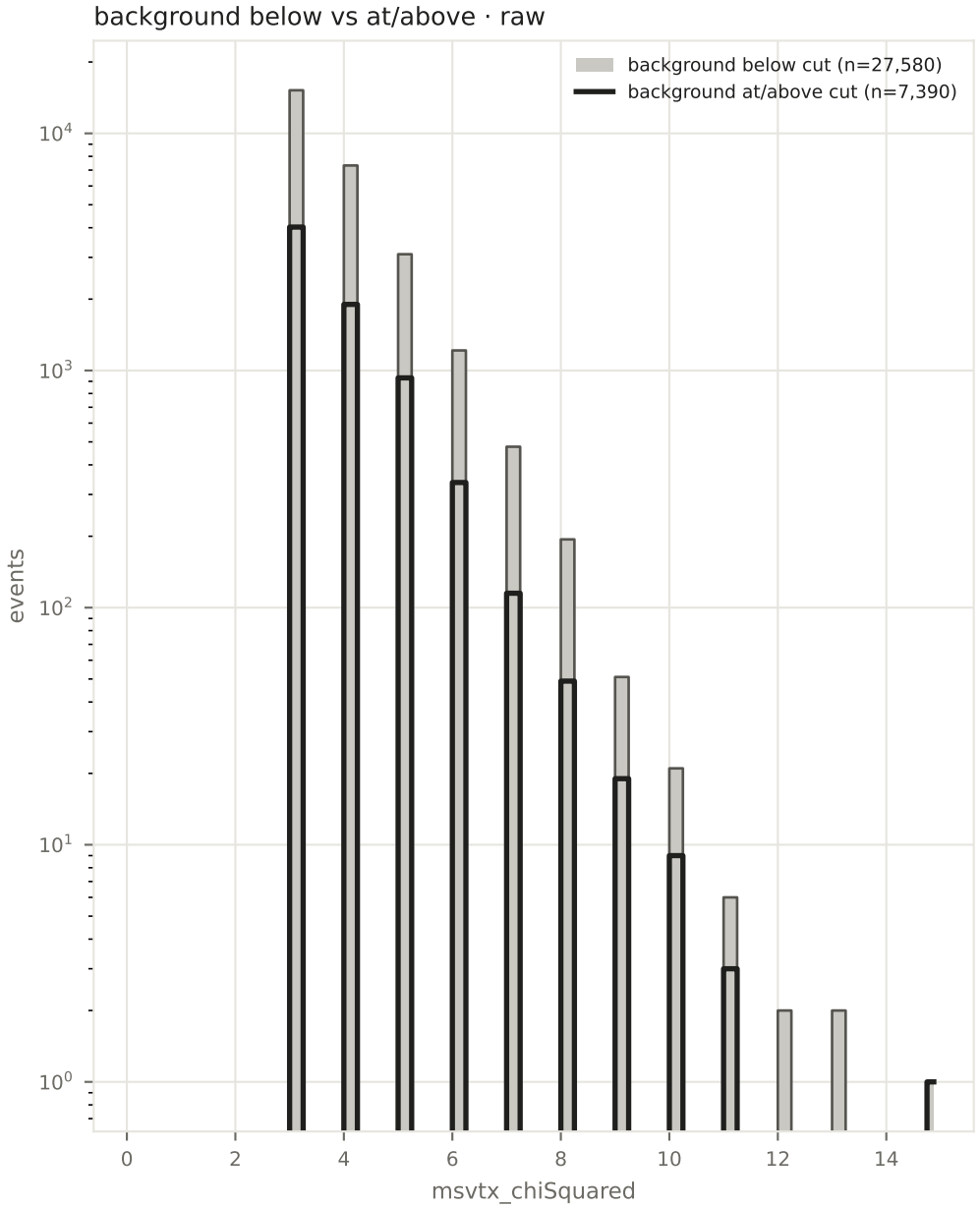


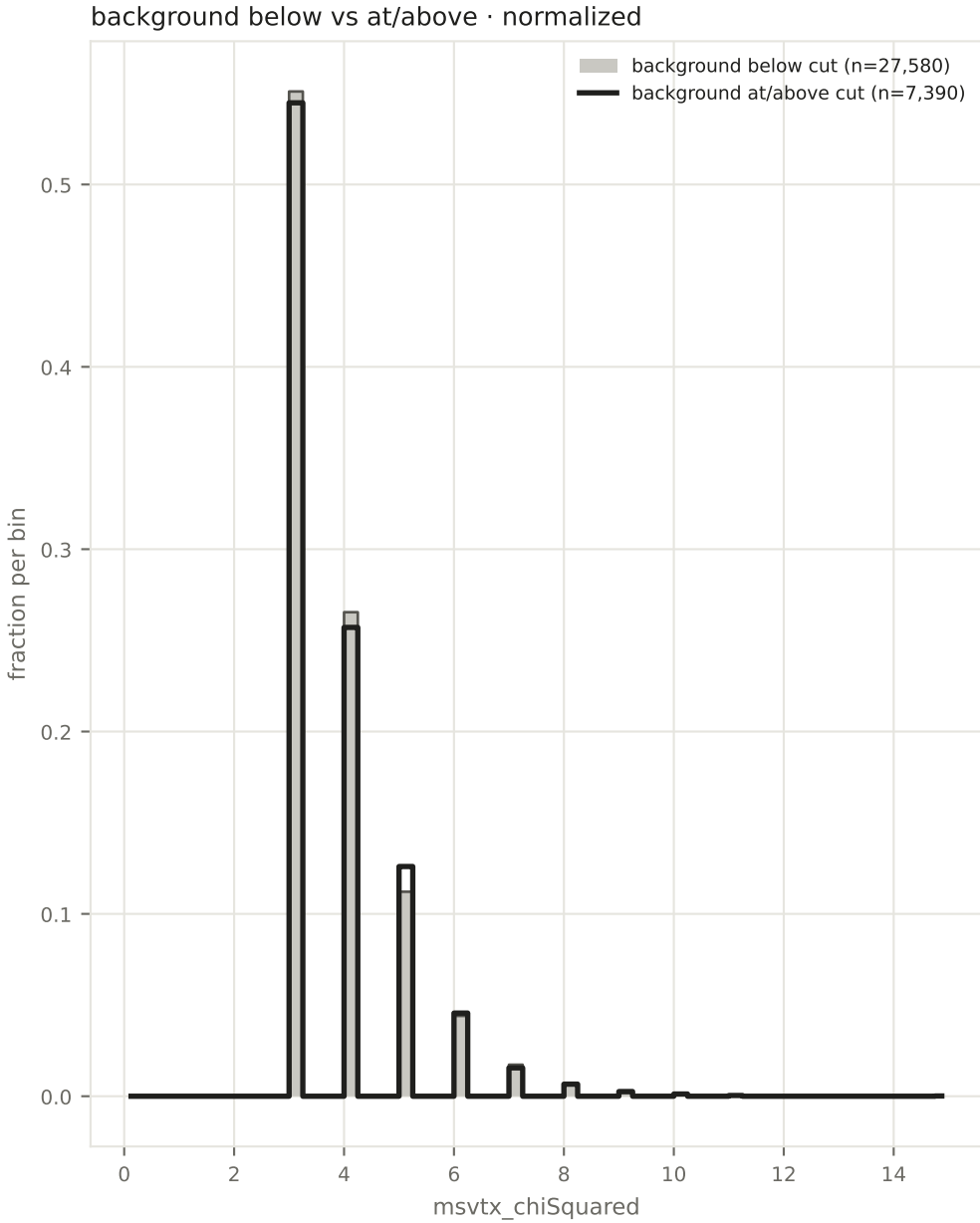
edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

region: ENDCAP · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 16 / 169

D(below vs above) 0.015 n below = 27,580 n above = 7,390 large D = background shape changes across the cut



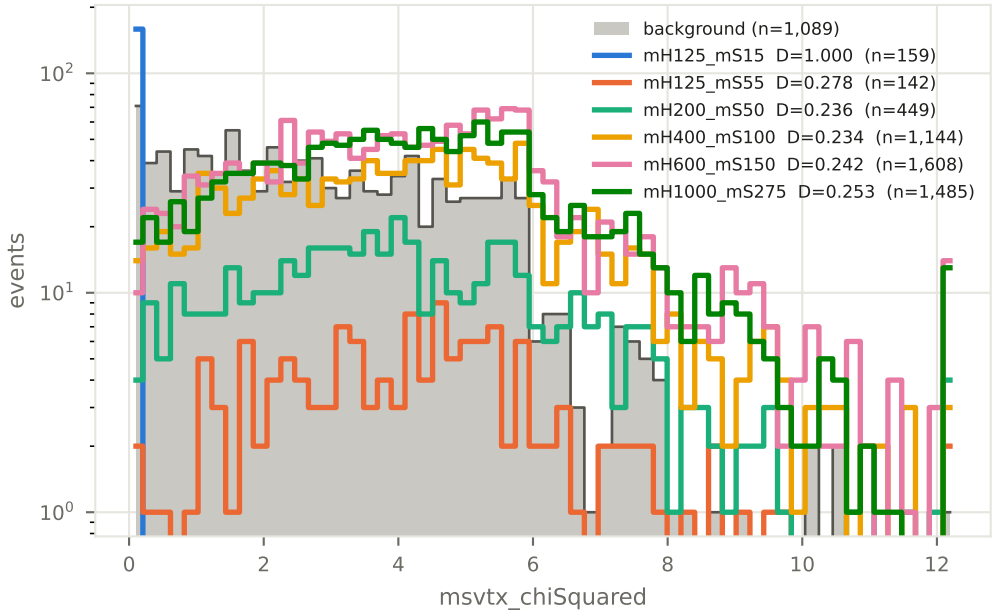


msvtx_chiSquared

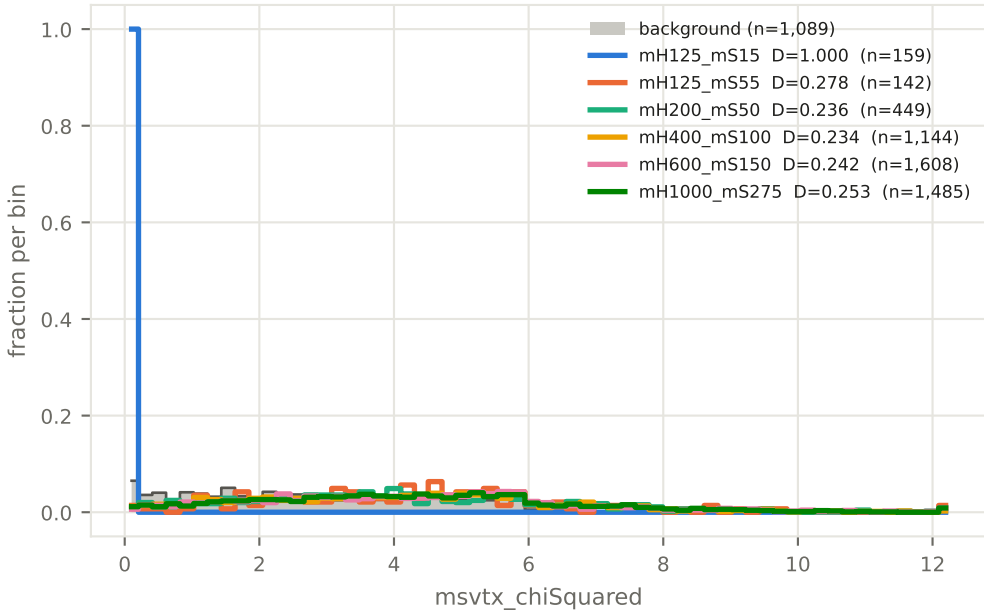
region: BARREL · selection: all (every event) · rank 16 / 169

D_equal 0.167 D_pooled 0.220 min 0.234 (mH400_mS100) max 1.000 (mH125_mS15) bkg n = 1,089 sig n = 4,987

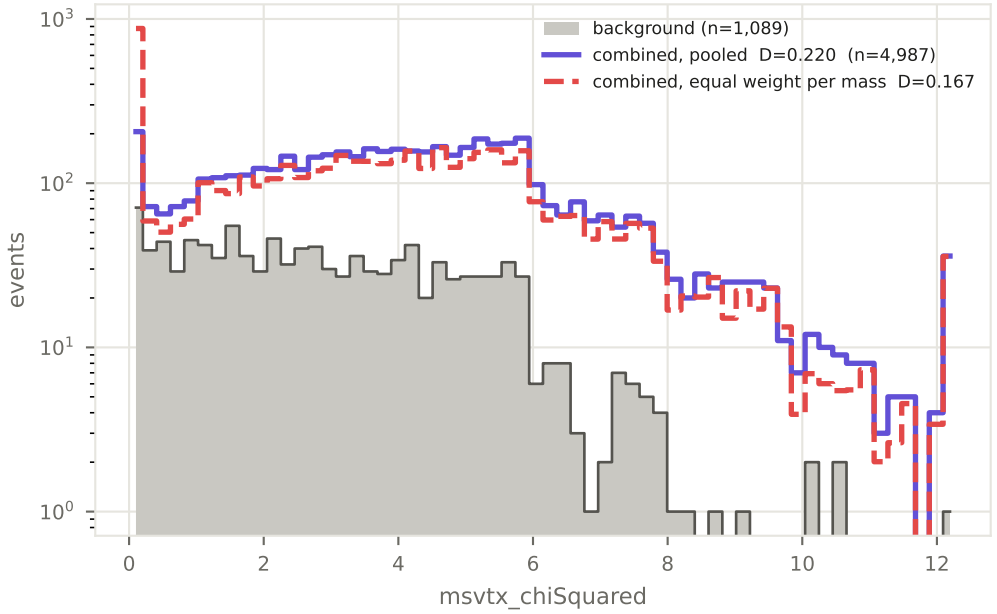
each mass point vs background · raw



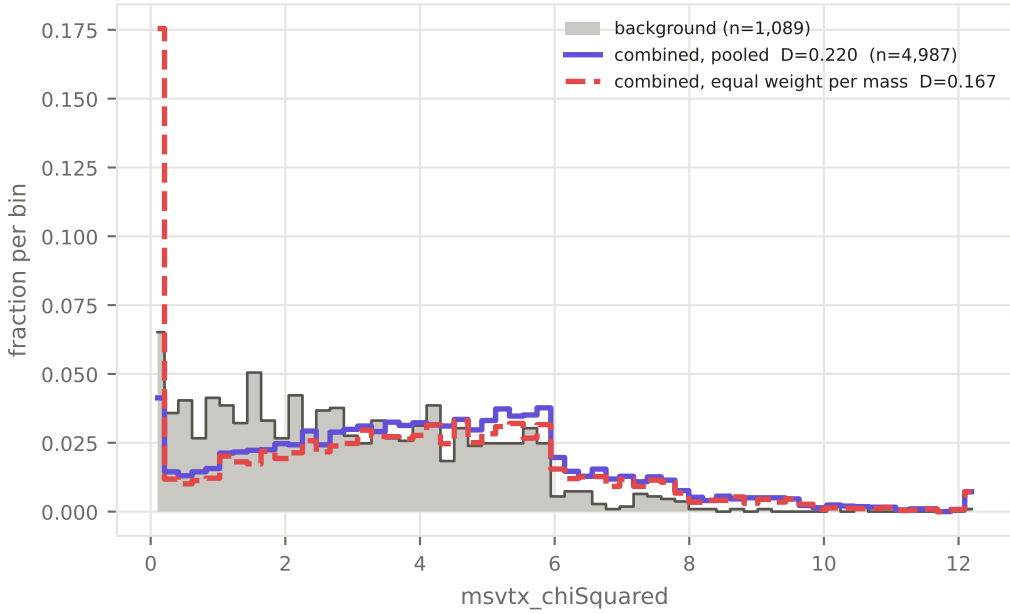
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



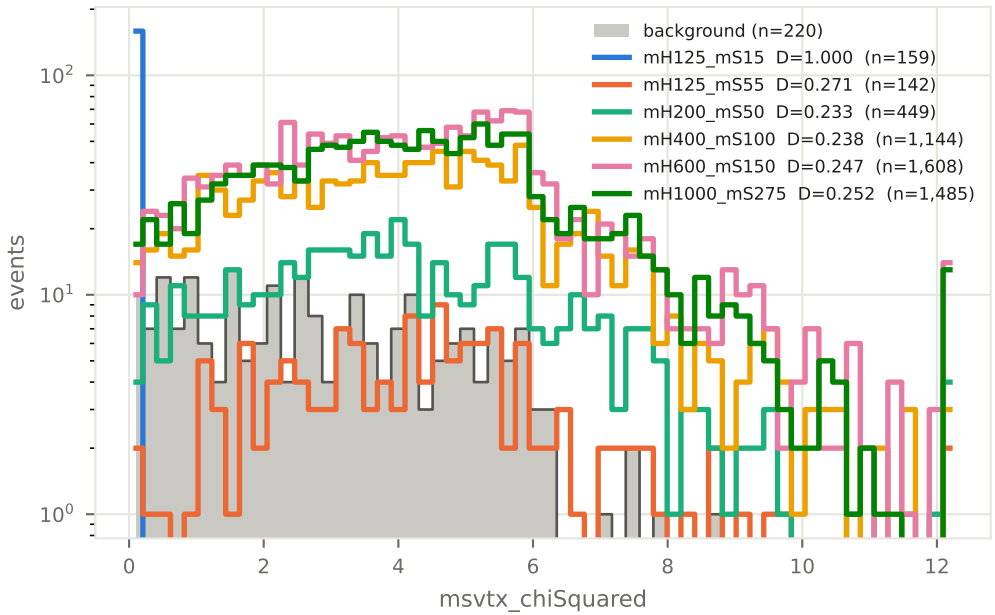
edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

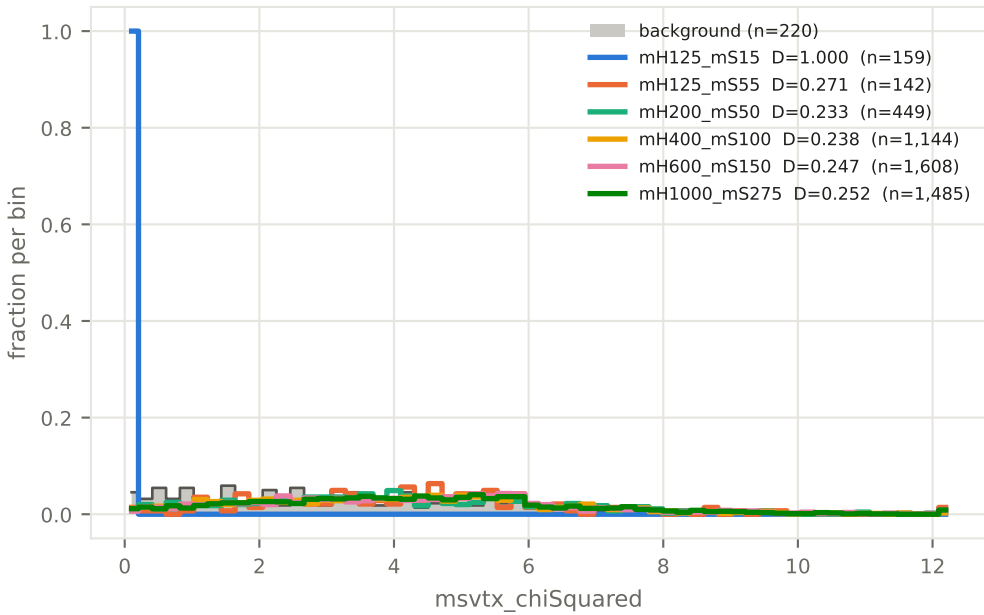
region: BARREL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 16 / 169

D_equal 0.167 D_pooled 0.221 min 0.233 (mH200_mS50) max 1.000 (mH125_mS15) bkg n = 220 sig n = 4,987

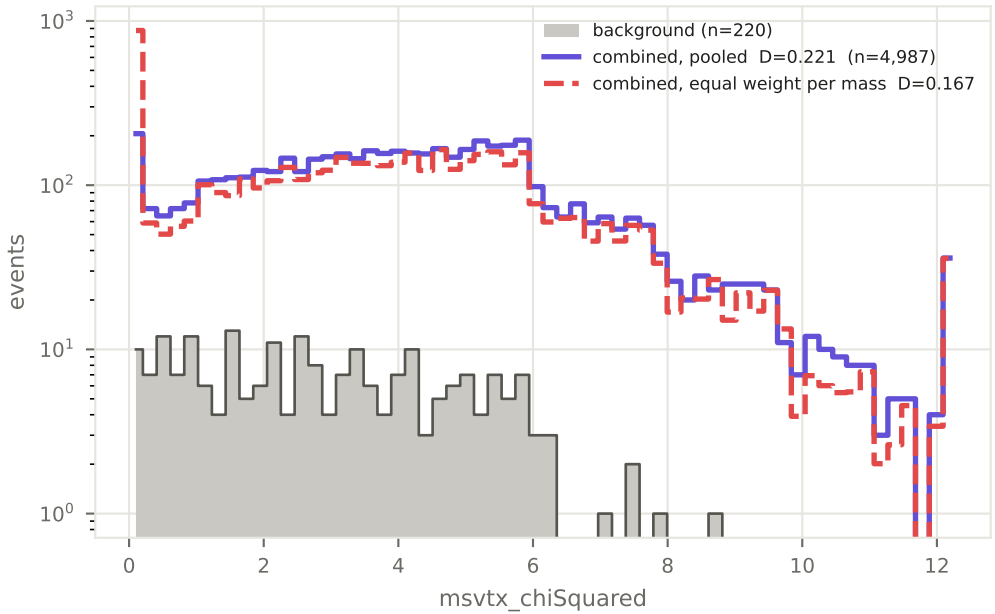
each mass point vs background · raw



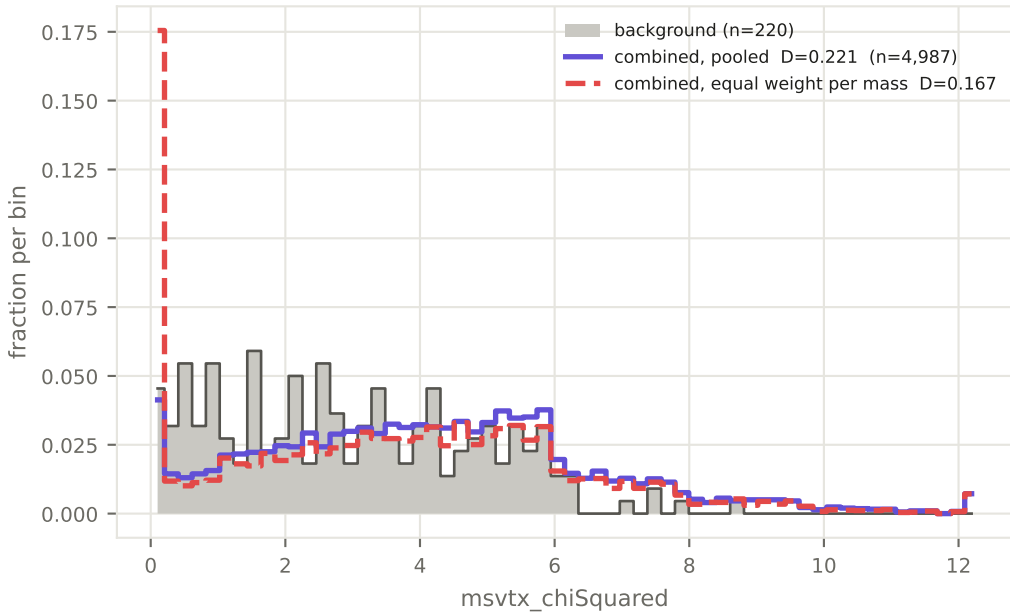
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized

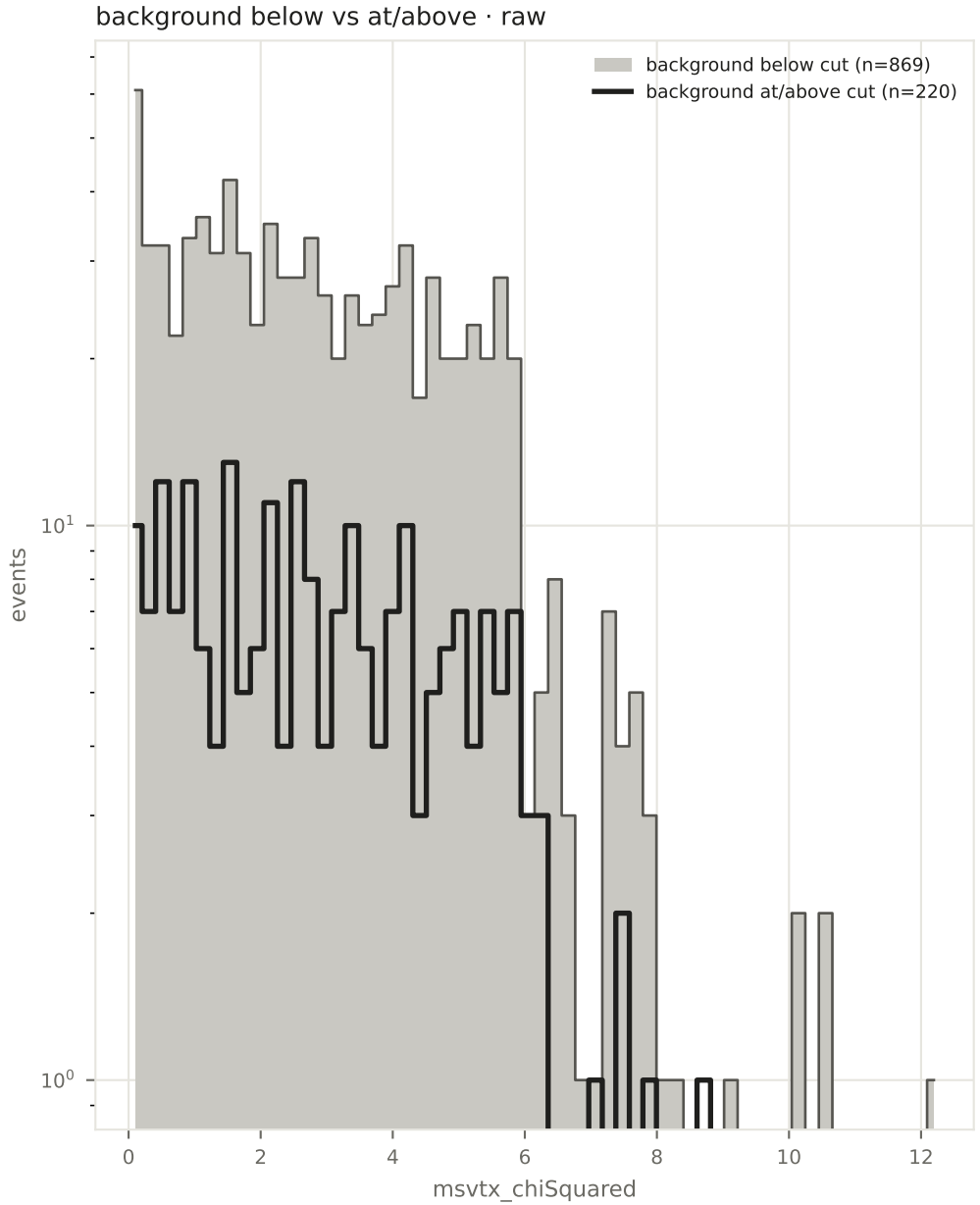


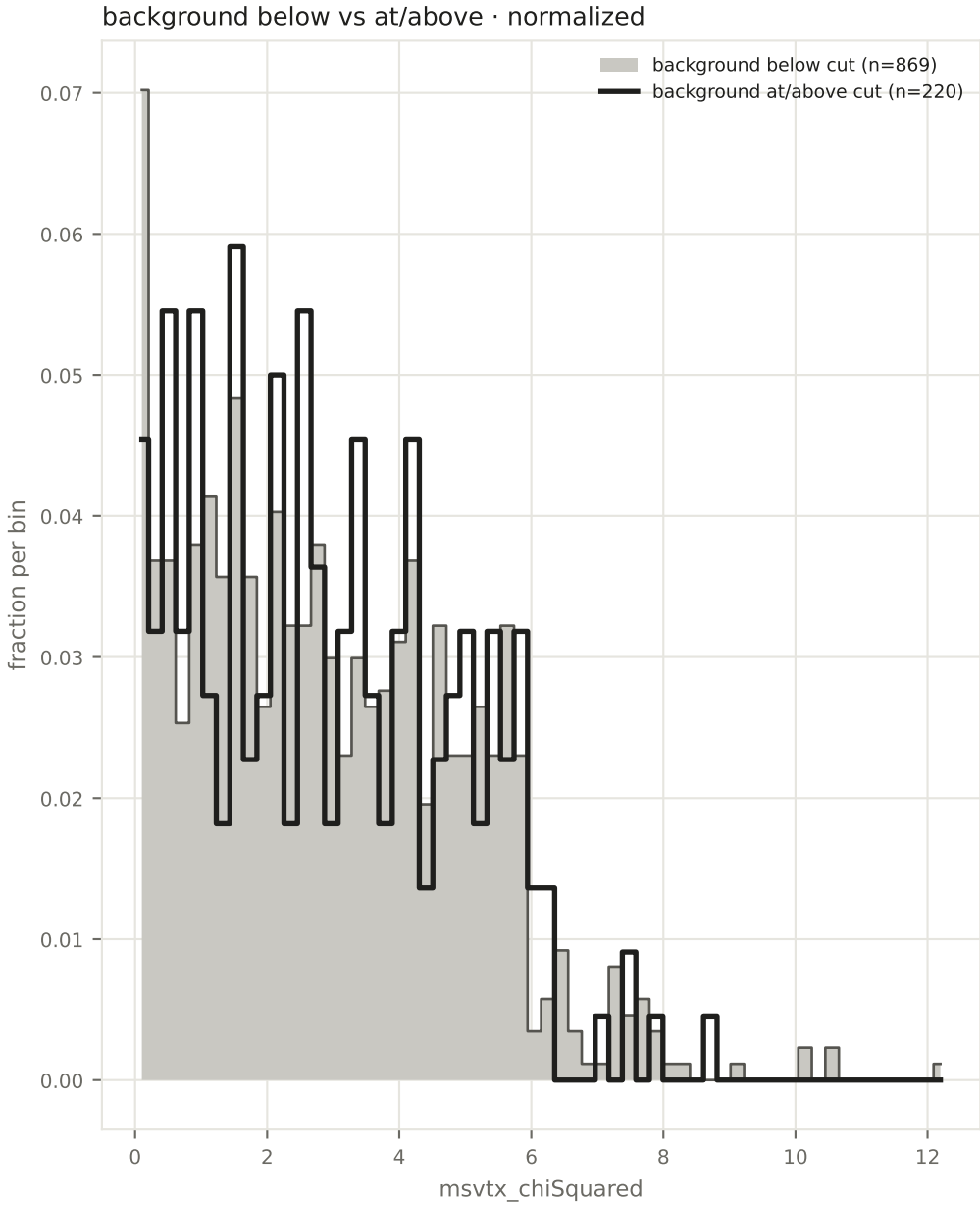
edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

region: BARREL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 16 / 169

D(below vs above) 0.035 n below = 869 n above = 220 large D = background shape changes across the cut



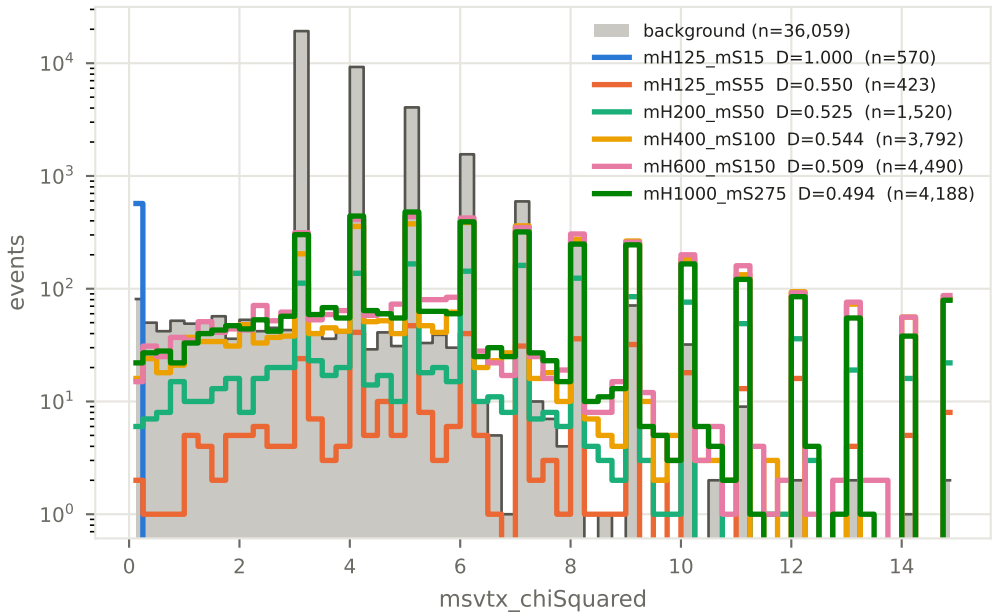


msvtx_chiSquared

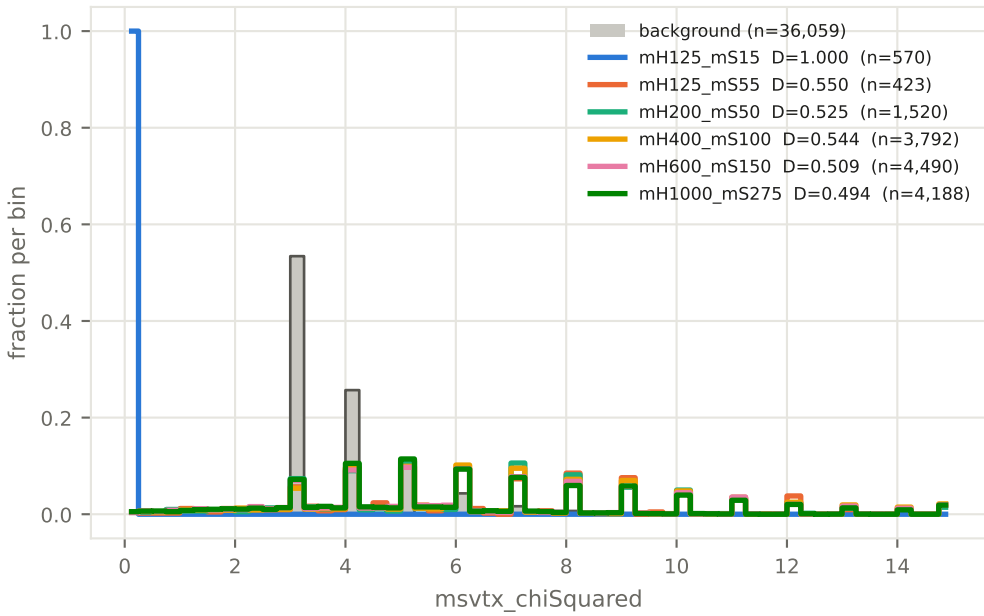
region: ALL · selection: all (every event) · rank 16 / 169

D_equal 0.405 D_pooled 0.490 min 0.494 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 36,059 sig n = 14,983

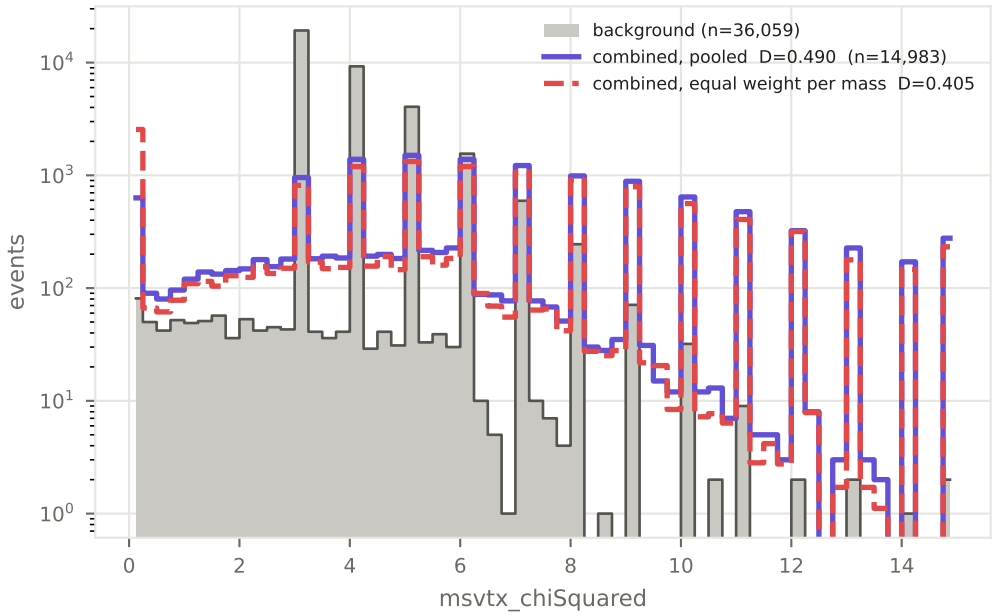
each mass point vs background · raw



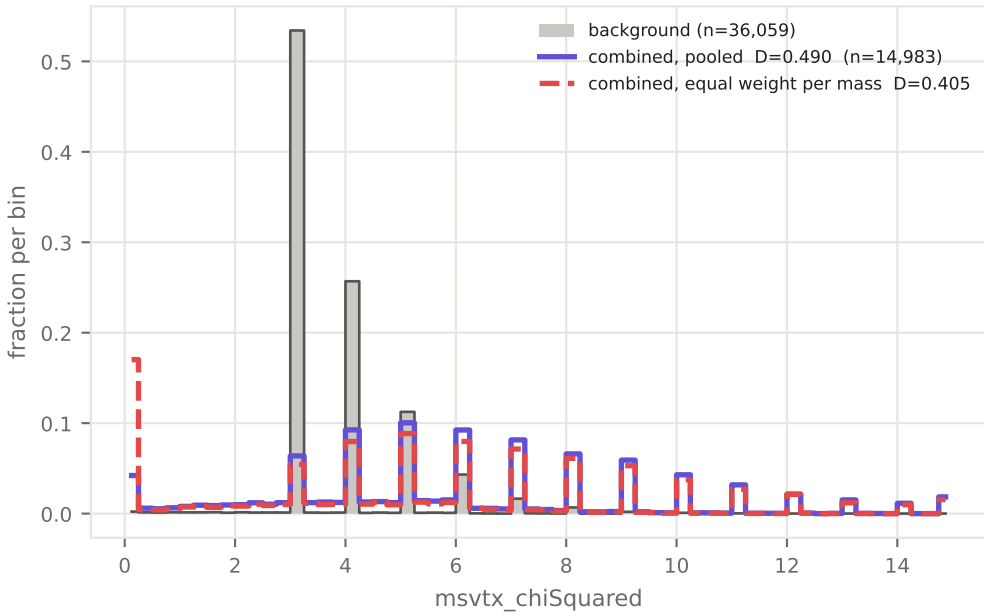
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



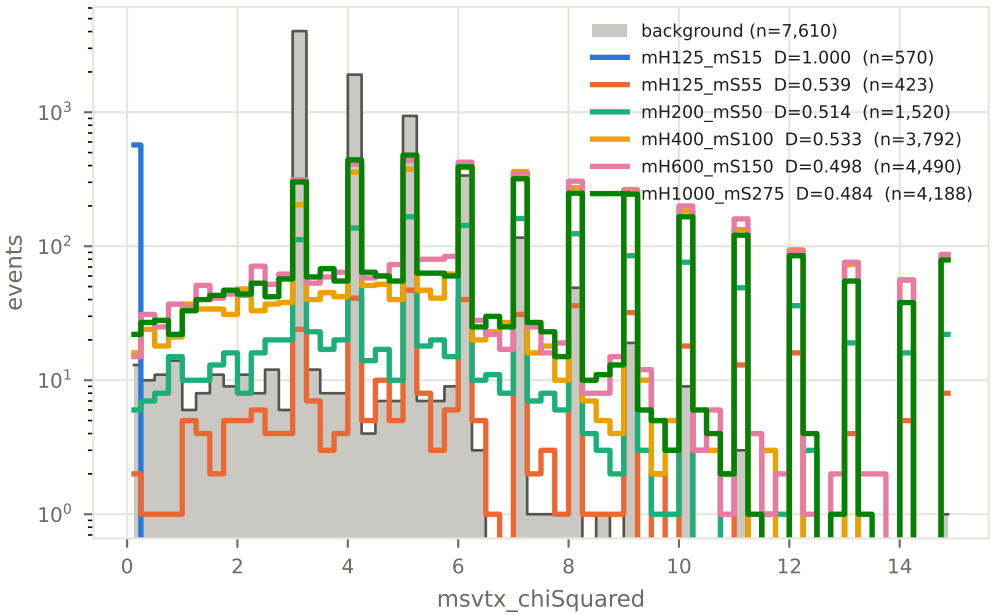
edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

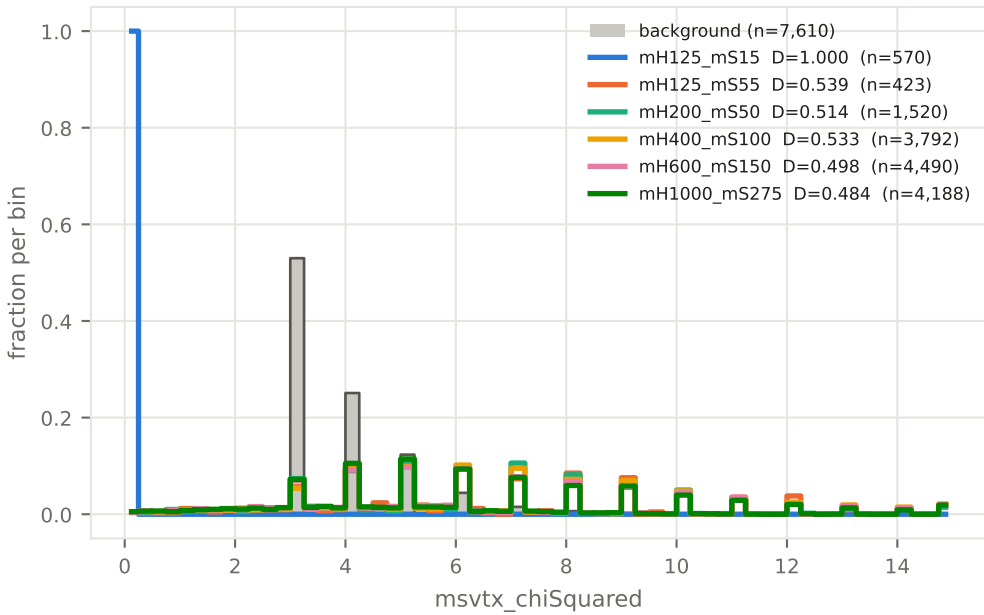
region: ALL · selection: ht40 (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 16 / 169

D_equal 0.399 D_pooled 0.479 min 0.484 (mH1000_mS275) max 1.000 (mH125_mS15) bkg n = 7,610 sig n = 14,983

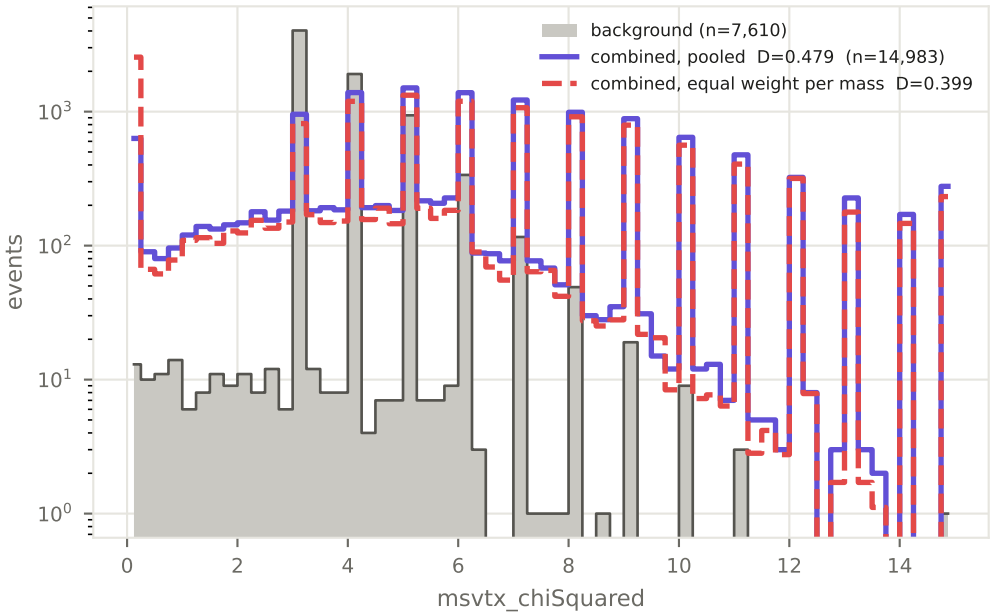
each mass point vs background · raw



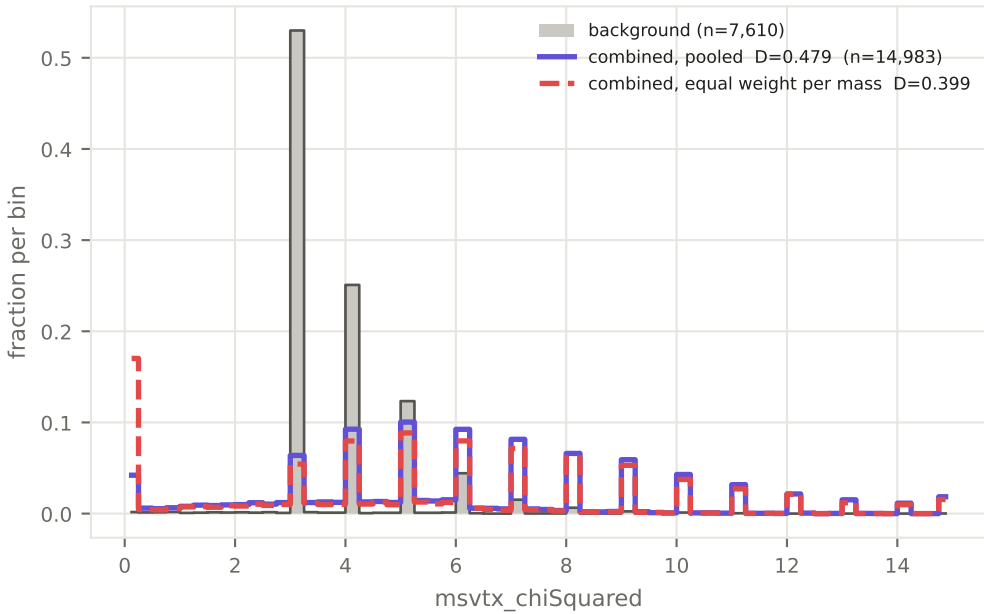
each mass point vs background · normalized



combined signal vs background · raw



combined signal vs background · normalized



edge bins include under/overflow · NaN and -999 removed

msvtx_chiSquared

region: ALL · background only: below vs at/above the cut (htmissScalar_NOSYS >= 40 GeV, signal and background) · rank 16 / 169

D(below vs above) 0.015 n below = 28,449 n above = 7,610 large D = background shape changes across the cut

